

The Hand That Takes

*Hittite $d\bar{a}$ - as pre-PIE $*de + a$, the Anatolian directive univerbation, and the within-Hittite chronology*

Paper 4 of the *der-* project. Builds on Chavis & Claude (2026a), (2026b), and (2026c).

[Working paper v5 — for scholarly review]

Chavis, J.M. and Claude, A.I.

Abstract

The first three papers of the *der-* project reconstructed pre-PIE $*de$ - “hand, open palm” and decomposed both the body-part noun $*der$ - and the give-verb $*deh_3$ - into the same bare hand-element extended by different morphemes. Paper 1 set out the morphological skeleton of the framework. Paper 2 developed the contrastive suffix $*-tero$ - and the numeral $*duo-/dui$ - “two” as derivational siblings on the same root. Paper 3 developed the directional particle $*de/do$, the locative $*-r$ -, and the give-verb $*deh_3$ - as bare $*de$ - + $*-h_3$ -, with the Greek root-aorist 2pl imperative contrast (δότε vs. στῆτε) as direct discriminating evidence for the bimorphemic decomposition.

The present paper carries the framework into Anatolian. We argue that Hittite $d\bar{a}$ - “to take” preserves a fossilized univerbation of pre-PIE $*de$ - with a postposed directional element $*a$ “toward (the speaker),” contracted by hiatus to long $*\bar{a}$, at a stage before the verbal extension $*-h_3$ - had been added. The take/give polarity thereby resolves into a structural opposition: bare $*de$ - “extend (the hand)” yields the giving direction in post-Anatolian PIE, while $*de$ - extended by directional $*a$ “toward the speaker” yields the taking direction preserved in Hittite $d\bar{a}$ -.

The argument’s centerpiece is the Anatolian X+a parallel: $*de + a > d\bar{a}$ - fits a documented Anatolian morphological type rather than constituting an isolated speculation. The Hittite Set 1 directional system — $anda < *en + a$, $katta < *k\bar{m}t + a$, $par\bar{a} < *pro + a$, $\bar{s}ar\bar{a} < *uper + a$ — is a productive class of fixed adverbs and preverbs formed by univerbation of a local element with the directional particle $*a$. The standard view treats $d\bar{a}$ - as an isolated irregular $\bar{h}$ -verb; the framework treats it as a regular member of a productive class.

The framework’s chronological architecture aligns with three independently published positions. Norbruis (2021) argues that the original meaning of $*deh_3$ - was “to take” (with “to give” as a post-Anatolian innovation). Bauhaus (2019) reconstructs a paradigmatic opposition between locative $*-r$ and directive $*-o$ on the same nominal roots, with the directive remaining productive into Anatolian — exactly the asymmetry the framework reconstructs. The Paper 1 / Paper 2

architecture identifies three derivational outcomes on **de-* that Anatolian preserves (bare **de-* in the directional particle; **de-* + **-u-* in the numeral; **de* + *a* > **dā-* in the take-verb) plus a fourth that is post-Anatolian (the **-h₃-* extension giving the give-verb).

The Luwian *lā-* problem is the framework’s biggest single liability and is engaged substantively in §9. The standard view derives Luwian *lā-* from PA **dh₃-* via fricativization. We argue that this derivation is ad hoc — the proposed sound change **d’vocalism-* > Luwian *l-* has no general parallel in Luwian phonology; the only documented Luwian sound change involving *l* is rhotacism, which goes the wrong direction. The framework does not propose a positive Luwian-internal etymology; the question remains genuinely open. The within-Hittite chronology of *-šten(i)* (Kloekhorst 2007) is a soft cost: the simplex *dā-* patterns with laryngeal-final stems, mildly against the framework’s prediction. The mitigating positive is the OS hapax *pe-tí-iš-te-en* in compound *peda-*, preserving archaic features the simplex has lost — exactly the fossilized-univerbation residue the framework predicts.

The conditional probability is assessed at 30–42%, a downward revision reflecting honest accounting of the Luwian *lā-* problem and the within-Hittite chronology cost, balanced against the X+a parallel as constructive positive argument and the architectural convergence of the four-paper framework.

1. Introduction

This is the fourth paper in the *der-* project. The first three established that pre-PIE had a body-part etymon meaning “hand, open palm”; that this etymon underlies a network of lexical, morphological, and verbal reflexes whose unity the standard handbooks have not recognized; and that the same etymon admits multiple parallel derivations producing the inherited IE system. Paper 1 §2.2 sets out the morphological skeleton: the body-part noun **de-r-* and the heteroclitc gift-noun as the lexical branch; the contrastive suffix **de-* + **-r-* + **-o-* > **(t)ero-* as the suffix branch; the numeral **de-* + **-u-* + **-o-/i-* > **du-o-/du-i-* as the numeral branch; the give-verb **de-* + **-h₃-* > **deh₃-* as the verbal branch; and (the contribution of the present paper) the take-verb **de* + *a* > **dā-* as the directive-univerbation branch.

Two facts about Anatolian, taken together, demand engagement. The first is well known: Hittite *dā-* / cognate Luwian *lā-* reflects the same etymon as the give-verbs of the rest of the family but means “to take,” not “to give.” The second is less often emphasized: Hittite preserves a productive synchronic system in which the bare *da-* stem combines with directional preverbs to specify deictic orientation, with *u-* “hither” yielding *uda-* “to bring here” and *pe-* “thither” yielding *pēda-* “to send away” (Kloekhorst 2008: 928–930; Hoffner & Melchert 2008: §20.7). The construction-type the framework hypothesizes — directional reversal of the bare hand-verb by an attached element — is therefore attested in

synchronic Hittite on the same root.

The thesis of this paper is that bare **dā-* “to take” is itself the residue of a similar but earlier construction, in which a postposed directional element **a* “toward the speaker” attached to the bare hand-element **de-* and underwent univerbation by hiatus contraction **e + a > *ā* in pre-Anatolian. The verbal extension **h₃-* of Paper 3 §6 is, on this view, a post-Anatolian innovation: a derivational suffix added to bare **de-* to create **deh₃-* yielding the give-verbs of the other branches. Anatolian split off before that extension was added. The paper is organized as follows: §2 sets out the standard reconstruction and the synchronic Hittite *u-/pe-* system; §3 develops the X+a parallel as the central positive argument; §4 presents the specific proposal and its chronological architecture; §5 addresses the within-Hittite chronology; §6 develops the semantic support from Norbruis (2021); §7 covers Boley (2007) and the CAPERE iconographic evidence; §8 sets out the PIE postpositional syntax context; §9 engages the Luwian *lā-* problem; §§10–13 close with future research, remaining problems, conclusion, and probability.

2. The standard reconstruction of Hittite *dā-* and the *u-/pe-* system

§2.1. Kloekhorst’s paradigm

Kloekhorst’s (2008: 928–930) entry on Hittite *dā-/d-* “to take” gives the canonical Old Hittite paradigm. The paradigm shows stem alternation between long-*ā* forms in the singular and short-vowel forms in the plural:

Form	Hittite	Notes
1sg pres.	<i>dāḫḫe / dāḫḫi</i>	strong stem <i>dā-</i>
2sg pres.	<i>dātti</i>	strong stem <i>dā-</i>
3sg pres.	<i>dāi</i>	strong stem <i>dā-</i>
1pl pres.	<i>tumēni, dāweni</i>	weak stem, /u/ or /aw/ vocalism
2pl pres.	<i>dattēni</i>	weak stem
3pl pres.	<i>danzi</i>	weak stem, /a/ vocalism

Kloekhorst derives this paradigm from PIE **deh₃-* with regular ablaut: the strong stem **dóh₃-* gives long *ā* via the regular Anatolian outcome of **ō < *oh₃*; the zero grade **dh₃-* gives /a/ via regular laryngeal vocalization in *danzi < *dh₃-énti*; the /u/ in *tumēni* comes from regular vocalization of zero-grade **dh₃-uéni*. One observation from Kloekhorst is directly relevant: the compound verbs *uda-* “to bring here” and *pēda-* “to bring away,” formed with the indissociable prefixes *u-* and *pe-*, preserve “forms that are more archaic, e.g. *petumen* and *utumen* vs. *dāwen*, *petišten* vs. *dāttēn*, *peter* and *uter* vs. *dāer*” (Kloekhorst 2008: 930). The paradigm of *dā-* is demonstrably innovative in its plural; the older state, accessible through the compounds, had a distinct stem alternation.

Melchert (1994: 89) reconstructs Proto-Anatolian **dō-* for the take-verb and identifies the long **ō* as the regular outcome that becomes Hittite long *ā*. His §4.1.6.3.1 documents that long vowels in Anatolian can derive from contracted vowel-plus-laryngeal or vowel-plus-vowel sequences — a principle directly relevant to the present proposal.

§2.2. The Hittite *u-/pe-* prefix system

Hoffner and Melchert (2008: §20.7) describe the Hittite *u-/pe-* preverb system synchronically: “In Hittite only two truly inseparable preverbs exist: *u-* ‘here’ and *pe-* ‘there’, attested in the pairs: *uda-* ‘to carry here’ and *peda-* ‘to carry away’; *uiya-* ‘to send here’ and *peya-* ‘to send away’; *unna-* ‘to drive here’ and *penna-* ‘to drive away’; *uwate-* ‘to lead here’ and *pehute-* ‘to lead away’.”

Two features bear on the framework. *First*, the construction-type the framework hypothesizes — directional reversal of the take-verb by an attached element — is real and productive in attested Anatolian. The objection that pre-PIE **de-* + *directional *a* is morphosyntactically unattested loses force when the same directional-reversal construction is documented for the same root in attested Hittite. *Second*, the long *ā* of bare *dā-* contrasts with the short vowel of the prefixed forms *uda-* and *pēda-*. Under the standard reconstruction this contrast follows from o-grade vs. zero-grade ablaut. Under the present proposal it falls out differently: the prefixed compounds preserve the bare stem attached to a prefixed directional, while the simplex preserves the fossilized univerbation **de + a > *dā* with the long vowel as the trace of the contracted directional.

3. The Anatolian X+a directional univerbation as positive structural argument

The framework’s claim that Hittite *dā-* derives from a pre-Anatolian univerbation **de + a* fits a documented Anatolian morphological type. The type is the productive Hittite Set 1 directional system, in which a local element is followed by the directional particle **-a* and contracts into a fixed adverb or preverb. The framework’s **de + a > dā-* is one instance of this type; it is not a one-off proposal. This section is the framework’s central positive structural argument.

§3.1. The Set 1 / Set 2 paradigmatic contrast

Hoffner and Melchert (2008: §20.3) describe the synchronic Hittite paradigmatic contrast between two paired sets of local adverbs:

Set 1 (“Place to Which”): *anda* “into,” *āppa* “back,” *katta* “down,” *parā* “out, forth,” *šarā* “upward.” Set 1 ends in *-a* and marks direction-toward. **Set 2 (“Place Where”):** *andan* “within,” *āppan* “behind,” *kattan* “below,” *pēran* “in front,” *šer* “above.” Set 2 ends in *-an* and marks location-at. The formal alternation *-a : -an*

maps consistently onto the directional/locative semantic contrast. The Set 1 forms transparently decompose as a local element followed by the directional **-a*: **en + a > anda*, **kmt + a > katta*, **pro + a > parā*, **uper + a > šarā*.

The directional **-a* is the same morpheme as the Hittite allative case ending *-a* attested productively in nominal forms: *aruna* “to the sea” (allative) vs. *aruni* “at the sea” (locative); *nepisa* “to heaven” vs. *nepisi* “in heaven.” Across both the local-adverb system and the nominal case system, *-a* marks direction-toward — the directional element the present paper proposes for the underlying component of Hittite *dā-*.

§3.2. **de + a > dā-* as one instance of the type

The framework’s **de + a > dā-* is structurally identical to **kmt + a > katta*. In each case, a local element is followed by the directional **-a* and contracts into a fixed form by hiatus. The contraction **kmt-a > katta* gives a geminated *tt* by regular vocalization of **m̥* before **t*; the contraction **de-a > dā* gives a long *ā* by regular hiatus contraction of **e + a*. Both contractions are documented in Anatolian phonology; both produce fossilized adverbs/preverbs that have lost their original morpheme boundary.

What is structurally distinctive about **de + a* — and what produced its surface as a verb rather than as an adverb/preverb — is that the local element **de-* was, alongside its directional function, also a verbal stem in the polyfunctional pre-PIE morphology of Paper 1. The univerbation of a verbal **de-* with the directional **a* yielded a fossilized verbal stem rather than a fossilized adverb: the take-verb **dā-*, deictically reversed from the bare give-direction **de-* by the postposed “toward (oneself)” element. This is the framework’s central positive structural argument: the X+a parallel converts an isolated proposal into one instance of a documented Anatolian morphological type.

§3.3. The hiatus contraction **e + a > *ā*

The contraction **e + a > *ā* needs phonological motivation. The Anatolian X+a paradigm provides the cross-form support: *parā* and *šarā* show long *ā* from analogous structures (**pro + a*, **uper + a* with metathesis or related developments). The nominal allative *aruna* “to the sea” preserves the same directional *-a* in nominal-case position. Melchert (1994: §4.1.6.3.1) documents that long vowels in Anatolian can derive from contracted vowel-plus-laryngeal or vowel-plus-vowel sequences across morpheme boundaries. The specific hiatus contraction proposed here is consistent with this general principle, though a fully worked-out sound rule for the specific environment remains a phonological loose end (§11).

§3.4. The standard view and what the X+a parallel adds

The standard view treats *dā-* as an isolated irregular *hi*-verb with no productive class membership. Kloekhorst’s reconstruction of *dā-* < **deh₃-* with regular o-grade laryngeal lengthening explains the long *ā* phonologically but does not place *dā-* within a productive Anatolian morphological type. The take/give polarity — the fact that Anatolian’s reflex of **deh₃-* means “take” while the post-Anatolian reflexes mean “give” — receives no structural source on the standard view.

The framework’s X+a reading places *dā-* within the productive Anatolian directional-univerbation type and provides a structural source for the take/give polarity: the polarity is not a semantic accident but a consequence of the directional **a* “toward (the speaker)” combining with the verbal **de-* “extend (the hand)” to give “extend toward oneself” = “take.” The bare **de-* without the directional yields the unmarked giving direction in post-Anatolian; **de* + *a* with the postposed directional yields the deictically reversed taking direction in Anatolian.

4. The proposal and its chronological architecture

§4.1. The framework’s specific proposal

In its earliest reconstructable stratum, pre-PIE was postpositional and OV (§8 below). At that stage, the bare hand-element **de-* appeared as a verbal stem, and a directional adverbial element **a* appeared postposed to it: **de-* + **a* → **de a* → **dā* (by hiatus contraction **ea* > **ā*). The semantics: bare **de-* “extend (the hand)” in the unmarked direction (= give, away from the speaker) plus directional **a* “toward the speaker” yields “extend toward oneself” = “take, receive.”

The construction is parallel to the synchronic Hittite *uda-* “to bring here” = bare verbal stem + directional “hither,” but with the directional element in postposed rather than prefixed position, reflecting the older morphosyntactic state. The proposal makes a specific prediction about the plural forms of the paradigm: the plural forms with /u/-vocalism (*tumēni*) and /a/-vocalism (*danzi*) reflect the bare verbal stem **d-* without the directional **a*, attached to inflectional endings whose vowels produced the surface forms. The long-*ā* of the singular subsequently spread into the plural in late Old Hittite — exactly the innovation Kloekhorst documents, and equally well-predicted by the present analysis as by the standard one.

§4.2. Bauhaus (2019): published external precedent

Bauhaus (2019: §5) reconstructs a paradigmatic opposition between locative **-r* and directive **-o* attached to the same nominal roots in early PIE. The pairs include **h₁upér* “on top” vs. **h₁upó* “to the top,” **k(e)nter* vs. **knto*, and **ndher* vs. **ndho*. Bauhaus summarizes: “all this strongly suggests that we are

dealing with a paradigmatic opposition ... whereas *-o* in its allative function could keep its productivity at least until Anatolian, *-r* has apparently ceased to be productive in all attested languages” (Bauhaus 2019: §5).

The framework reconstructs exactly this kind of paradigmatic opposition on the bare nominal root **de-* “hand”: locative **de-r-* (the body-part noun, Papers 1 and 3) versus directive **de + a* (the Anatolian take-verb, this paper). Bauhaus’s chronological observation additionally supports the Anatolian argument: the directive “*-o* in its allative function could keep its productivity at least until Anatolian,” while the locative **-r* “ceased to be productive in all attested languages.” Under the framework’s architecture, this is exactly the observed distribution: the locative member **de-r-* was fossilized as a body-part noun before the Anatolian split; the directive member **de + a* remained productive into Anatolian, where it contracted to **dā-* and was preserved as an active verb.

§4.3. The architectural convergence across Papers 1-4

The framework’s chronological architecture aligns with the morphological skeleton developed across all four papers. The pre-Anatolian chronology contains three derivational outcomes on **de-* that Anatolian preserves, plus a fourth that is post-Anatolian:

Stratum	Derivational pathway	Surface	Anatolian witness
Pre-Anatolian	bare <i>*de-</i>	directional particle <i>*de/do</i>	Hittite directional Set 1 <i>-a</i> ; Hittite allative <i>-a</i>
Pre-Anatolian	<i>*de- + *-u- + *-o-/-i-</i>	numeral <i>*du-o-/du-i-</i> “two”	Hittite <i>dāyu-</i> “two” (standard)
Pre-Anatolian	<i>*de- + *a</i> (directive univertation)	take-verb <i>*dā-</i>	Hittite <i>dā-</i> “to take” (this paper)
Post-Anatolian	<i>*de- + *-h₃-</i> (verbal extension)	give-verb <i>*deh₃-</i>	absent from Anatolian

The framework’s chronology is therefore not ad hoc but a coherent system in which Anatolian preserves multiple pre-extension derivational outputs of the same root, with a specific identifiable derivational pathway absent from Anatolian (the **-h₃-* verbal extension). The convergence with Norbruis (2021)’s independent semantic chronology and Bauhaus (2019)’s independently published locative-directive paradigmatic pairs is mutually reinforcing across three independent lines of evidence.

5. The within-Hittite chronology

§5.1. The three waves of replacement

Kloekhorst (2007) documents that the replacement of the original *hi*-conjugation 2pl ending *-šten(i)* by the *mi*-conjugation *-tten(i)* in Hittite happens in three chronological waves. Wave 1 (pre-Hittite) affects consonant-final stems. Wave 2

(pre-Hittite → Old Script, OS) affects laryngeal-final stems. Wave 3 (Middle Hittite → New Hittite) affects vowel-final stems. The simplex *dā-* “to take” already shows *-tten* in OS texts (*da-at-te-e-ni*, *da-a-at-te-en*) — it patterns with laryngeal-final stems that received the replacement in pre-Hittite times (Wave 2), not with vowel-final stems that received the replacement only in Middle-to-New Hittite (Wave 3).

§5.2. The cost: simplex *dā-* patterns with laryngeal-final

Under the framework’s analysis (where *dā-* derives from **de + a* by hiatus contraction, producing a vowel-final stem **dā-*), the simplex should pattern with the vowel-final class (Wave 3, replaced only in MH-NH) rather than the laryngeal-final class (Wave 2, replaced pre-Hittite to OS). The standard view, which derives *dā-* from **deh₃-* with a root-final laryngeal, predicts the laryngeal-final patterning directly. This is a real cost for the framework.

§5.3. The mitigating positive: *peda-* and *petišten*

Two mitigating considerations support the framework against this cost. *First*, the OS hapax *pe-tí-iš-te-en* in the compound verb *peda-* “to bring away” preserves two archaic features that the simplex *dā-* has lost: the *-šten* ending itself (the Wave 1/2 archaism) and a weak stem *ti-* not attested in the simplex paradigm. The compound therefore preserves an older paradigm shape than the simplex, exactly as the framework predicts for a fossilized univerbation: the simplex *dā-* has been leveled and lexicalized as a single stem, while the compounds preserve traces of an older state. Kloekhorst’s own observation — that the compounds show “forms that are more archaic” — is the published support for this reading.

Second, the framework’s hiatus contraction **e + a > *ā* produces a phonologically heavy syllable that may pattern morphologically with stems ending in a long vowel from laryngeal coloring (**eh₂ > *-aH > *-ā*). By the time of attested Hittite, the original morpheme boundary of **de + a* was no longer detectable; the verb behaved like a unitary stem with a long *ā* and was classified morphologically with the laryngeal-final group by analogical accommodation.

§5.4. Net assessment

The *-šten(i)* chronology is a soft cost rather than a fatal one. The compound *peda-* preserving *petišten* with the archaic weak stem *ti-* is a positive datum: the compound preserves features the simplex has lost, exactly as the framework predicts. The simplex’s patterning with the laryngeal-final class is mildly against the framework’s prediction but is accommodatable through the analogical-class argument. The standard view fits this datum more cleanly, but the cleaner fit comes at the cost of treating the compound’s preservation of *petišten* and weak stem *ti-* as an unmotivated archaism rather than as expected residue of a

fossilization process.

6. Norbruis (2021): independent semantic support

Norbruis (2021: 243–249), in his Leiden dissertation chapter on the etymology of **deh₃-*, argues independently for a chronology closely aligned with the morphosyntactic chronology proposed here. His conclusion: “In this chapter it is argued that the verb **deh₃-* did not mean ‘to give’ in PIE, but ‘to take’, as in Anatolian. Although the verb did not survive in any evident way in Tocharian and Germanic, the semantic shift to ‘to give’ is likely to be an innovation of post-Anatolian IE.”

Norbruis works through every parallel adduced for a give > take semantic development and shows they actually support the opposite direction. He reconstructs a semantic chain: “to take” > (with Goal or Beneficiary constituent) “to bring, convey, deliver” > (when the constituent was animate) “to hand, to give.” The intermediate stage — physical delivery of concrete objects to animate recipients — is not only reconstructed: the Prestino inscription (northern Italy, 5th–6th c. BCE) provides direct epigraphic attestation. Eska (2004) analyzes the Prestino formula as describing the physical delivery of concrete objects (seats) to animate divine recipients — exactly Norbruis’s reconstructed intermediate stage, not yet the fully abstract “give” of later Celtic usage.

Norbruis’s chronology aligns directly with the morphosyntactic framework proposed here. Anatolian split off not only before the semantic shift but also, on the framework’s account, before the **-h₃-* verbal extension was added. The two analyses are decoupleable: Norbruis’s semantic argument is fully independent of his formal reconstruction, and the framework’s formal proposal can stand or fall independently of the semantic chronology. We take the coincidence of formal and semantic chronology as mutually reinforcing.

7. Boley (2007) and the iconographic evidence from CAPERE

§7.1. Boley as historiographic precursor

The proposal that **deh₃-* derives from a directional particle is not unprecedented. Boley (2007) previously proposed deriving the give/take root from “a directionally ambivalent particle reconstructed on the basis of Latin *dē* ‘from’ and Greek *-δε* ‘to’” (as reported by Norbruis 2021: 245 n.9). The same Greek allative clitic *-δε* that Paper 3 adduces for **de-* appears in Boley’s earlier proposal.

Norbruis (2021: 245 n.9) dismisses Boley’s specific arguments as unsound, and we agree with much of his criticism. The notion of a “directionally ambivalent particle” with both “from” and “to” senses simultaneously is not well motivated.

What survives the criticism is the structural insight: the give/take verb of PIE derives from a directional particle. The present paper offers a more constrained version: rather than a single ambivalent particle, we propose a directional-toward element **a* which combined with the bare hand-element **de-* to yield the take-meaning, while the bare element alone in post-Anatolian developed the give-meaning through the **-h₃-* extension. The directional contrast is specific, the morphology more precisely specified, and the chronology supplied by Norbruis (2021).

§7.2. Iconographic evidence: Anatolian hieroglyph CAPERE

Anatolian hieroglyphic sign L 41 (Laroche numeration) is iconographically a “taking” or “grasping” hand. It functions both as the logogram CAPERE “to take” — used to write words for taking, most commonly the Luwian verb *lā-* — and as a syllabogram with phonetic value /tā/ (Rieken 2008; Yakubovich 2008). Yakubovich (2008) derives the syllabic value /tā/ acrophonically from Hittite *dā-* “to take” in the bilingual Hittite-Luwian environment of script development.

Under the present framework — where Hittite *dā-* derives from pre-PIE **de-* (the bare hand-element) + postposed directional **a* — the hand iconography of L 41 is not a coincidence of universal cognition but an iconographic preservation of the verb’s etymological source. The Anatolian script-creators chose a hand to depict *dā-* because the verb was, in its deep history, a hand-derivation. A competing analysis is offered by Kloekhorst (2019), who derives the value [ḏa] of L 41 from Luwian *lā-* reconstructed as PA **dh₃-* with the laryngeal preserved as fricativization. This re-analysis depends on the contested phonological framework engaged in §9. Yakubovich’s senior acrophonic analysis does not depend on contested mechanisms; we treat it as the cleaner basis for interpreting the iconographic evidence. The iconographic evidence is corroborative, not decisive.

8. PIE postpositional syntax: the early-stratum context

The framework’s proposal requires that pre-PIE allowed postposed adverbial directional elements to appear after verbal stems. This is the unmarked configuration for the earliest reconstructable layers of Indo-European syntax. Lehmann’s (1974) *Proto-Indo-European Syntax* argued on typological grounds that PIE was an OV language whose unmarked phrasal pattern required postposed adpositions and modifiers. Friedrich (1975) refined the typology. Four direct attestations support the early postpositional state.

Hittite is uniformly postpositional in its adpositional system: all Hittite adpositions are postpositions in nominal phrases (*parni anda* “into the house,” with *anda* after the dative-locative *parni*). As the earliest-attested Indo-European branch, Anatolian’s postpositional pattern is a strong indicator of the PIE state.

Tocharian, the second-earliest branch to split off under most current chronologies, is also predominantly postpositional. Vedic Sanskrit preserves substantial postpositional flexibility. Latin preserves postpositional archaisms in fossilized form: *mēcum* “with me,” *tēcum* “with you,” *nōbiscum* and *vōbiscum* all have *cum* postposed to the personal pronoun.

Homeric Greek shows the relevant change in real time. In Homer the so-called “preverbs” remain separable from the verb (tmesis: κατὰ δάκρυα λείβων “shedding tears down,” with κατά free-standing rather than fused into Classical καταλείβων). By Classical Greek these elements have univerbated with the verb stem. The construction we propose for the bare hand-element — **de-* + postposed directional **a* > fossilized **dā-* — is a special case of this documented PIE pathway.

9. The Luwian *lā-* problem

The Luwian *lā-* problem is the framework’s biggest single liability. The standard view (Kloekhorst 2019, following Norbruis 2021) derives Luwian *lā-* “to take” from PA **dh₃-* via fricativization. We argue that this derivation is ad hoc and depends on a contested phonological framework. The framework does not propose a positive Luwian-internal etymology; the question of where Luwian *lā-* comes from remains genuinely open.

§9.1. The standard reconstruction

The standard reconstruction — Kloekhorst (2008, 2013), Norbruis (2021) — derives the verb from PA **doH-* / **dH-* with the laryngeal **h₃* preserved as glottalization of the initial stop. The same laryngeal is held to give Luwian *lā-* “to take” via a post-PA development **d’- > Luwian l-*, with the laryngeal preserved as fricativization on the initial dental and subsequently developing to a lateral. Kloekhorst (2019) makes this argument explicitly for Luwian *lā-* and uses it to explain the value of Anatolian hieroglyph L 41 (§7.2).

§9.2. Three considerations against the standard view’s derivation

First, the proposed sound change **d’- > Luwian /l-/* has no independent parallels elsewhere in Luwian or Anatolian phonology. The documented Luwian sound change involving *l* is rhotacism: in Iron Age Hieroglyphic Luwian, intervocalic *d, l, n* → *r* (e.g. **īdi* → *īri* “he gets,” *wala-* → *wara-* “die”; Yakubovich and Shatskov). This change goes the wrong direction for Kloekhorst’s derivation and operates intervocalically, not word-initially. Word-initial **d-* normally remains *d-* in Luwian; there is no general sound law producing *l-* from word-initial **d-*. The derivation is therefore ad hoc — invented for this single etymology.

Second, the underlying glottalization framework on which the Kloekhorst derivation depends is contested. Melchert (2015) argues explicitly: “There is no evidence for word-initial $*h_1$ preserved as [ʔ] in Hittite and Luvian, contra Kloekhorst (2004, 2006a: 77–81 and 2008: 205 and passim) and Simon (2010 and 2013).” Sideltsev and Molina (2021) raise independent methodological objections to Kloekhorst’s broader prosodic-phonological framework.

Third, $*h_3$ has the weakest independent evidence base of the three PIE laryngeals. $*h_1$ leaves demonstrable e-coloring; $*h_2$ leaves clear a-coloring and compensatory-lengthening patterns; $*h_3$ reduces primarily to o-coloring in Greek and the contested Anatolian glottalization argument. Kloekhorst’s derivation of Luwian $lā-$ stands at the most vulnerable point of laryngeal theory.

§9.3. What the framework can and cannot claim

The framework’s position: the standard view’s derivation has no general sound law to support it, depends on a contested glottalization framework, and rests on the most vulnerable point of laryngeal theory. The framework does not have a positive Luwian-internal etymology to offer; the question of where Luwian $lā-$ comes from is genuinely open. What the framework can claim is that the standard view’s derivation does not constitute decisive evidence against the framework’s pre-Anatolian chronology.

The standard analysis has the advantage of formal economy within the laryngeal-theoretic framework and derives Hittite $dā-$ / Luwian $lā-$ from a single PIE root with established sound laws. It has the disadvantage that the take/give polarity receives no structural source. The framework’s derivation is structurally motivated by the X+a parallel, evidentially grounded by the Norbruis chronology and Bauhaus’s locative/directive paradigmatic pairs, and internally consistent with the morphological skeleton of Papers 1–3. What the framework provides, and the standard account does not, is a structural source for the take/give polarity.

10. Future research

Other potential fossilized univerbations in Hittite ʔi-verbs. If pre-Anatolian had a productive $*X + a$ univerbation pattern that affected verbs, $*de + a$ should not be the only example. Whether other Hittite ʔi-verbs with isolated long $\bar{a}$ might be analyzable as fossilized univerbations of bare verbal stems with postposed directional elements is a question for further work.

Palaic and the minor Anatolian languages. The framework’s pre-Anatolian chronology should make predictions about what Palaic, Lycian, Lydian, etc. should show. What does the comparative Anatolian evidence look like beyond Hittite and Luwian? This is the natural extension that would test the framework’s chronology against a wider Anatolian sample.

The Luwian $\bar{l}a$ - problem. Whether a positive Luwian-internal etymology can be developed for $\bar{l}a$ - — independent of the contested $*dh_3$ - derivation — is the framework’s biggest single liability and the most consequential extension. The framework registers the question as genuinely open.

11. Remaining problems

Three problems deserve flagging; none is fatal, but two are real.

First, the Luwian $\bar{l}a$ - problem (§9). The framework does not propose a positive Luwian-internal etymology. The mitigating consideration is that the standard view’s derivation is itself ad hoc and depends on a contested phonological framework, but the framework’s failure to provide an alternative is a real gap.

Second, the within-Hittite chronology of $-\acute{s}ten(i)$ (§5). The simplex $d\bar{a}$ - patterns with laryngeal-final stems rather than vowel-final stems, mildly against the framework’s prediction. The mitigating consideration is the OS hapax *petišten* in compound *peda*- preserving archaic features the simplex has lost — exactly the fossilized-univerbation residue the framework predicts.

Third, the hiatus contraction $*e + a > *\bar{a}$ (§3.3) requires sharper specification. The Anatolian X+a paradigm provides analogous cases, but the specific contraction across a morpheme boundary in verbal-stem position needs a fully worked-out sound rule that the present paper does not provide. This is a phonological loose end rather than a structural problem.

12. Conclusion

Hittite $d\bar{a}$ - “to take” preserves a fossilized univerbation of pre-PIE $*de$ - (the polyfunctional hand-element of Paper 1) with a postposed directional element $*a$ “toward (the speaker),” contracted by hiatus to long $*\bar{a}$ in pre-Anatolian. The verbal extension $*-h_3$ - of Paper 3 §6 is post-Anatolian: a derivational suffix added to bare $*de$ - to create $*deh_3$ - yielding the give-verbs of the other branches. Anatolian split off before that extension was added.

The framework’s central positive structural argument is the Anatolian X+a parallel (§3): $*de + a > d\bar{a}$ - fits a documented Anatolian morphological type alongside $anda < *en + a$, $katta < *k\acute{m}t + a$, $par\bar{a} < *pro + a$, $\acute{s}ar\bar{a} < *uper + a$. The standard view treats $d\bar{a}$ - as an isolated irregular hi -verb; the framework treats it as a regular member of a productive class.

The framework’s chronological architecture aligns with three independently published positions. Norbruis (2021) reconstructs “to take” as the original meaning of $*deh_3$ - on independent semantic grounds. Bauhaus (2019) reconstructs a paradigmatic opposition between locative $*-r$ and directive $*-o$ on the same

nominal roots, with the directive remaining productive into Anatolian — exactly the asymmetry the framework reconstructs. The four-paper architecture identifies three pre-Anatolian derivational outcomes on **de-* that Anatolian preserves plus a fourth that is post-Anatolian, organized as a coherent system. The Luwian *lā-* problem remains genuinely open: the framework does not propose a positive etymology, and the standard view’s derivation is ad hoc. The take/give polarity has a structural source on the framework and none on the standard view.

13. Probability estimate

Conditional on Papers 1, 2, and 3 being correct, the probability that the substantive thesis of this paper is correct in its basic outline is approximately 30–42%. Two costs drive the downward adjustment from earlier estimates.

The Luwian lā- problem (§9) is the dominant downward pressure. The framework does not propose a positive Luwian-internal etymology; the question is genuinely open. The mitigating consideration is that the standard view’s derivation is itself ad hoc and depends on a contested phonological framework.

The within-Hittite chronology cost (§5) is a softer but real downward pressure. The simplex *dā-* patterns with laryngeal-final stems rather than vowel-final stems, mildly against the framework’s prediction. The mitigating positive is the *peda-* compound’s preservation of *petišten* and weak stem *ti-*, exactly the fossilized-univerbation residue the framework predicts.

Three considerations support the upper end of the range. *The X+a parallel as constructive positive argument (§3)*: the framework’s central claim is now that **de + a > dā-* is one instance of a documented Anatolian morphological type, not an isolated speculation. *The architectural convergence (§4.3)*: the framework’s pre-Anatolian chronology contains three derivational outcomes that Anatolian preserves plus a fourth that is post-Anatolian, organized as a coherent system. *The Norbruis (2021) and Bauhaus (2019) convergences*: two independently published positions support the framework’s chronology.

The marginal probability — unconditioned, integrating over the probability that Papers 1, 2, and 3 are also correct — is approximately: 30–40% (Paper 1) × 35–45% (Paper 2) × 35–45% (Paper 3) × 30–42% (Paper 4) ≈ 1.1–3.4%. Much of the evidence supports multiple strata simultaneously rather than each stratum independently; the honest marginal range for the framework as a whole is in the 1–4% range. We continue to regard the central claim as more likely wrong than right at the marginal level, and to regard it as interesting enough, and the evidential structure dense enough, that publication for examination by other scholars is warranted.

14. Authorship and contributions

Chavis's contributions. The hypothesis that Hittite *dā-* “to take” preserves an earlier construction in which the bare hand-element **de-* was extended by a postposed directional element originates with Chavis. All decisions about scope, framing, and inclusion are Chavis's. The decision to foreground the X+a parallel as the load-bearing positive argument — making the Anatolian morphological type the constructive center of the case rather than background context — is Chavis's. The decision to engage the Luwian *lā-* problem substantively rather than rely on the iconographic argument alone is Chavis's. The decision to add §5 on the within-Hittite chronology is Chavis's, made after identifying the *-šten(i)* data as genuinely informative about the framework's prediction structure. The hypothesis that PIE postpositional syntax dissolves the central morphosyntactic objection to the framework's construction is Chavis's. All fundamental decisions about scope, framing, and direction across all four papers rest with Chavis, who supplies all intellectual continuity through systematic transcript management.

Claude's contributions. Claude reviewed prior-thread material on the Luwian *lā-* problem (Kloekhorst 2019, Melchert 2015, Sideltsev & Molina 2021) and on the within-Hittite chronology (Kloekhorst 2007 on *-šten(i)*); searched for documented Luwian sound changes involving *l* (finding rhotacism as the only general change, going the wrong direction for Kloekhorst's derivation); drafted the §3 foregrounding of the X+a parallel as positive structural argument; drafted the §5 engagement of the within-Hittite chronology with the *peda-* / *petišten* compound as mitigating positive evidence; drafted the §9 substantive engagement of the Luwian *lā-* problem; drafted the §4.3 architectural convergence; and integrated new sections with preserved material in §§2, 6, 7, 8. Claude additionally read Kloekhorst (2008, 2013, 2019), Hoffner & Melchert (2008), Norbruis (2021), Bauhaus (2019), Boley (2007), Yakubovich (2008), Rieken (2008), Melchert (1994, 2015, 2017), Lehmann (1974), and Friedrich (1975) throughout the project's arc. Claude drafted all prose sections from Chavis's structural and substantive direction. The probability estimate in §13 is Claude's own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

Select references

- Bauhaus, F. 2019. “The Proto-Indo-European Suffix **-r* Revisited.” In Goldstein, R. et al. (eds.), *Sequencæ, Svb-Sequencæ and Singularities in Phonology*, 13–26. Leiden: Brill.
- Boley, J. 2007. *dā-, pai-* “give” and the relationship of Anatolian to PIE. In Groddek, D. & Zorman, M. (eds.), *Tabularia Hethaeorum*, 83–90. Wiesbaden: Harrassowitz.

- Chavis, J.M. and Claude. 2026a. "The Hand That Gives." *LingBuzz* (v33).
- Chavis, J.M. and Claude. 2026b. "A Better Hand." *LingBuzz* (v10).
- Chavis, J.M. and Claude. 2026c. "The Bare Hand." *LingBuzz* (v9).
- Eska, J.F. 2004. "Continental Celtic." In Woodard, R.D. (ed.), *Cambridge Encyclopedia of the World's Ancient Languages*, 857–880. Cambridge University Press.
- Friedrich, P. 1975. *Proto-Indo-European Syntax: The Order of Meaningful Elements*. JIES Monograph 1. Butte, MT.
- Hawkins, J.D. 2000. *Corpus of Hieroglyphic Luwian Inscriptions*, Vol. I. Berlin: De Gruyter.
- Hoffner, H.A. and H.C. Melchert. 2008. *A Grammar of the Hittite Language. Part 1: Reference Grammar*. Winona Lake: Eisenbrauns.
- Jasanoff, J.H. 2003. *Hittite and the Indo-European Verb*. Oxford University Press.
- Kloekhorst, A. 2007. "The Hittite syllabification of PIE *CuR and *KuR." [On the -šten(i) > -tten(i) replacement chronology.]
- Kloekhorst, A. 2008. *Etymological Dictionary of the Hittite Inherited Lexicon*. Leiden: Brill. [Entry on dā-/d- "to take": pp. 928–930.]
- Kloekhorst, A. 2013. "The signs TA and DA in Old Hittite: evidence for a phonetic difference." *Altorientalische Forschungen* 40: 125–141.
- Kloekhorst, A. 2019. "The phonetic value of Anatolian hieroglyphic ⟨41⟩ (tà) and the development of *dh₃- in Luwian." *Kadmos* 58(1–2): 23–42.
- Kloekhorst, A. and T. Pronk (eds.). 2019. *The Precursors of Proto-Indo-European*. Leiden: Brill.
- Lehmann, W.P. 1974. *Proto-Indo-European Syntax*. Austin: University of Texas Press.
- Melchert, H.C. 1994. *Anatolian Historical Phonology*. Amsterdam: Rodopi.
- Melchert, H.C. 2015. "Hittite Historical Phonology after 100 Years (and after 20 years)." Manuscript, UCLA.
- Melchert, H.C. 2017. "An allative case in Proto-Indo-European?" In Hansen et al. (eds.), *Usque ad Radices*, 527–540. Copenhagen: Museum Tusculanum.
- Norbruis, S. 2021. *Indo-European Origins of Anatolian Morphology and Semantics*. LOT Dissertation 588. Amsterdam: LOT. [Chapter 7, pp. 243–249.]
- Payne, A. 2015. *Hieroglyphic Luwian: An Introduction with Original Texts*. 3rd ed. Wiesbaden: Harrassowitz.
- Rieken, E. 2008. "Die Zeichen ⟨ta⟩, ⟨tá⟩ und ⟨tà⟩ in den hieroglyphen-luwischen Inschriften der Nachgroßreichszeit." *Studi Micenei ed Egeo-Anatolici* 50: 637–649.
- Sideltsev, A. and M. Molina. 2021. Review of A. Kloekhorst, *Accent in Hittite. Babel und Bibel* 9: 259–278.
- Yakubovich, I. 2008. "Hittite-Luvian bilingualism and the development of Anatolian hieroglyphs." *Acta Linguistica Petropolitana* 4(1): 9–36.
- Yakubovich, I. and A. Shatskov. n.d. "Rhotacism in 1st-Millennium BC Anatolia. Comparative Luwian and Lydian Phonology."